

Chapter 3: Discrete Distributions

Probability and Statistics for Science and Engineering
with Examples in R

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(1) Random Variable (r.v.)

Definition: any function that assigns a numerical value to each possible outcome of an experiment

Example 3.1: X : # heads obtained in three tosses of a fair coin

Outcome	X
HHH	3
HHT	2
HTH	2
HTT	1
THH	2
THT	1
TTH	1
TTT	0

Value of X	Event
$X = 0$	$\{TTT\}$
$X = 1$	$\{HTT, THT, TTH\}$
$X = 2$	$\{HHT, HTH, THH\}$
$X = 3$	$\{HHH\}$

$$P(X = 0) = \frac{1}{8}$$

$$P(X = 1) = \frac{3}{8}$$

$$P(X = 2) = \frac{3}{8}$$

$$P(X = 3) = \frac{1}{8}$$

Types of Random Variables

- **Discrete Random Variables** have a countable number of possible values.
- **Continuous Random Variables** can take on any value in an interval and cannot be enumerated.

eg) X : # heads obtained in three tosses of a coin: discrete

X : amount of precipitation produced by a storm: continuous

(2) Probability Distribution

1. Probability distribution: $f(x) = P(X = x)$
2. $f(x) \geq 0$ for all x
3. $\sum_{\text{all } x} f(x) = 1$

Example 3.2: Which of the following is a probability distribution?

- $f(x) = \frac{x-1}{3}$ for $x = 0, 1, 2, 3$

$$f(0) = -\frac{1}{3} < 0, \text{ so not a probability distribution.}$$

- $f(x) = \frac{x^2}{12}$ for $x = 0, 1, 2, 3$

$$f(x) \geq 0 \text{ for all } x, \text{ but}$$

$$\sum_{x=0}^3 f(x) = 0 + \frac{1}{12} + \frac{4}{12} + \frac{9}{12} = \frac{14}{12} = \frac{7}{6} > 1$$

so not a probability distribution.

Example 3.3: Foreign made cars: 30%. Four cars are selected at random.
 X : number of foreign made cars. F: foreign made, D: domestic

$X = 0$	$X = 1$	$X = 2$	$X = 3$	$X = 4$
DDDD	DDDF	DDFF	DFFF	FFFF
	DDFD	DFDF	FDFF	
	DFDD	DFFD	FFDF	
	FDDD	FDFF	FFFD	
		FDFF		
		FFDD		

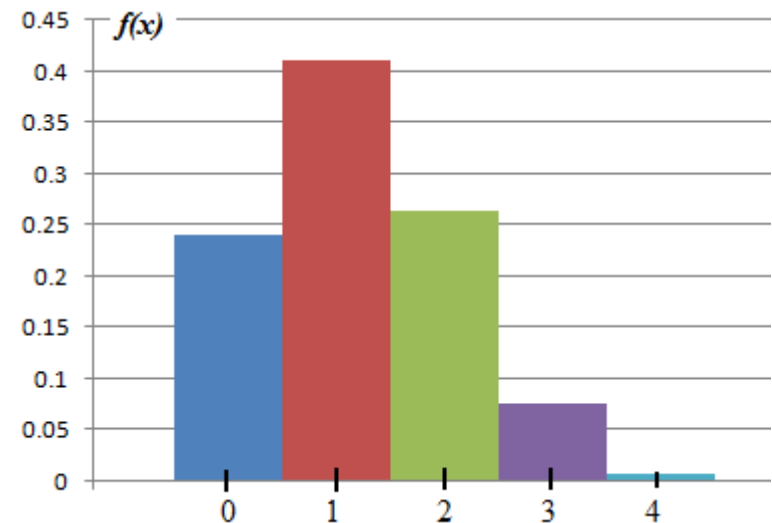
$$P(X = 0) = P(DDDD) = 0.7^4 = 0.2401$$

$$P(X = 1) = 4 \cdot 0.7^3 \cdot 0.3 = 0.4116$$

$$P(X = 2) = 6 \cdot 0.7^2 \cdot 0.3^2 = 0.2646$$

$$P(X = 3) = 4 \cdot 0.7 \cdot 0.3^3 = 0.0756$$

$$P(X = 4) = 0.3^4 = 0.0081$$



Probability Model: An assumed form of the probability distribution that describes the chance behavior for a r.v. X

Parameters: the relevant population quantities by which probabilities are expressed

Example 3.4: **Bernoulli Trial**

(a) Each trial yields one of two outcomes: Success (S) or Failure (F)

(b) $P(S) = P(X = 1) = p$; $P(F) = P(X = 0) = 1 - p$, or

$$f(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

(c) Each trial is independent.

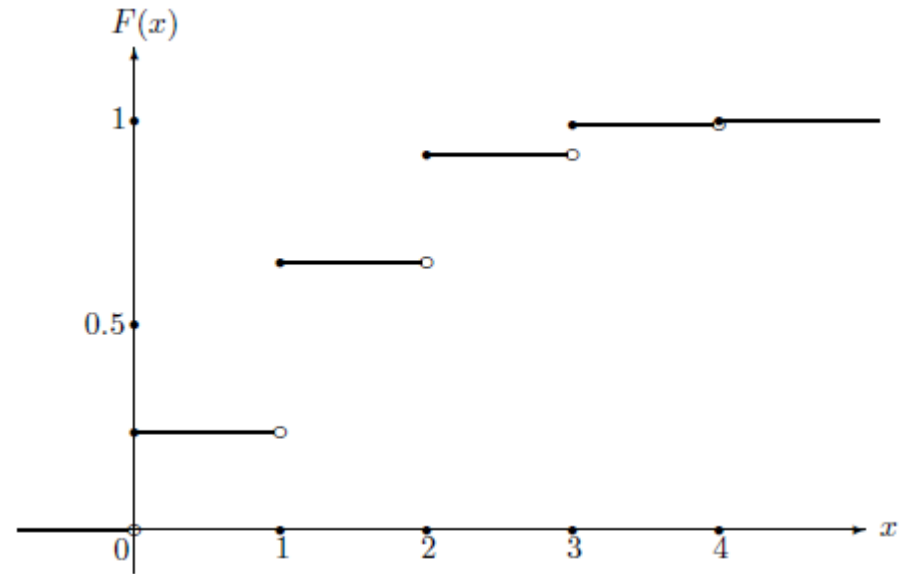
(p is the parameter)

Cumulative Distribution Function (CDF)

- CDF of a random variable: $F(x) = P(X \leq x) = \sum_{y \leq x} f(y)$
- For a discrete random variable:
 1. CDF has a “jump” at each possible value equal to the probability of that value.
 2. The graph of the cdf will be a step function.
 3. The graph increases from a minimum of 0 to a maximum of 1.

Example 3.3 (continued)

x	$f(x)$	$F(x) = P(X \leq x)$
0	0.2401	0.2401
1	0.4116	0.6517
2	0.2646	0.9163
3	0.0756	0.9919
4	0.0081	1



$$f(0) = 0.2401, f(1) = 0.4116, f(2) = 0.2646, f(3) = 0.0756, f(4) = 0.0081$$

$$F(0) = f(0) = 0.2401$$

$$F(1) = P(X \leq 1) = f(0) + f(1) = 0.2401 + 0.4116 = 0.6517$$

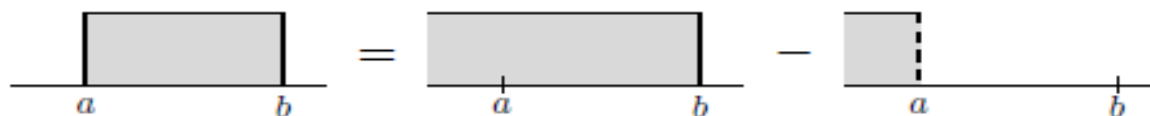
$$F(2) = P(X \leq 2) = f(0) + f(1) + f(2) = 0.9163$$

$$F(3) = P(X \leq 3) = f(0) + f(1) + f(2) + f(3) = 0.9919$$

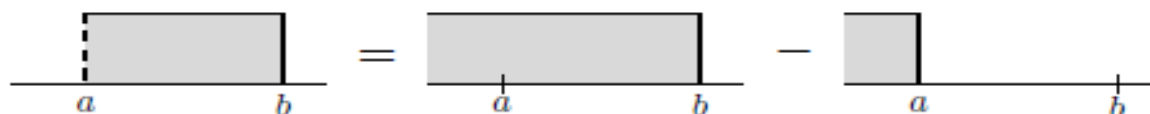
$$F(4) = P(X \leq 4) = 1$$

For $a < b$,

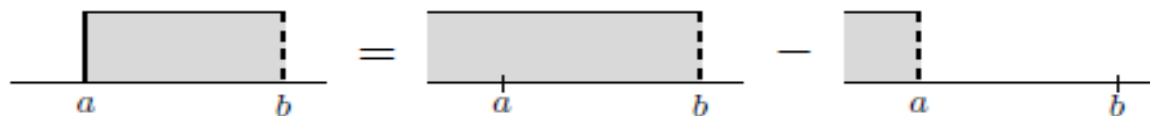
$$P(a \leq X \leq b) = P(X \leq b) - P(X < a) = F(b) - F(a^-)$$



$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a)$$



$$P(a \leq X < b) = P(X < b) - P(X < a) = F(b^-) - F(a^-)$$



$$P(a < X < b) = P(X < b) - P(X \leq a) = F(b^-) - F(a)$$



Example 3.5

x	$f(x)$	$F(x)$
0	0.1	0.1
1	0.2	0.3
2	0.3	0.6
3	0.2	0.8
4	0.2	1

$$P(1 \leq X \leq 3) = F(3) - F(0) = 0.8 - 0.1 = 0.7$$

$$P(1 < X \leq 3) = F(3) - F(1) = 0.8 - 0.3 = 0.5$$

$$P(1 \leq X < 3) = F(2) - F(0) = 0.6 - 0.1 = 0.5$$

$$P(1 < X < 3) = F(2) - F(1) = 0.6 - 0.3 = 0.3$$

(3) Mean and Variance of Discrete R.V.s

Definition: Mean (Expected value) of a discrete r.v. X :

$$E(X) = \mu = \sum_{\text{all } x} xf(x)$$

Example 3.6: What is the expected number of heads in three tosses of a fair coin?

X : # heads in three tosses of a fair coin

x	$f(x)$	$xf(x)$
0	$\frac{1}{8}$	0
1	$\frac{3}{8}$	$\frac{3}{8}$
2	$\frac{3}{8}$	$\frac{3}{4}$
3	$\frac{1}{8}$	$\frac{3}{8}$
Total	1	$\mu = 1.5$

Example 3.7: Mean of a Bernoulli r.v.:

$$f(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\mu = E(X) = \sum_x xf(x) = 0 \cdot f(0) + 1 \cdot f(1) = p$$

$$E(h(X)) = \sum_{\text{all } x} h(x)f(x)$$

Example 3.9: In flipping 3 balanced coins find $E(X^3 - X)$

x	$x^3 - x$	$f(x)$	$(x^3 - x)f(x)$
0	0	1/8	0
1	0	3/8	0
2	6	3/8	9/4
3	24	1/8	3
Total		1	21/4

$$E(X^3 - X) = 0 + 0 + \frac{9}{4} + 3 = \frac{21}{4} = 5.25$$

Variance of a discrete r.v. X :

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = \sum_{\text{all } x} (x - \mu)^2 f(x)$$

$$\text{sd}(X) = \sigma = \sqrt{\text{Var}(X)}$$

$$\text{Alternatively } \text{Var}(X) = E(X^2) - \mu^2 = \sum_{\text{all } x} x^2 f(x) - \left[\sum_{\text{all } x} x f(x) \right]^2$$

Proof:

$$\begin{aligned} \text{Var}(X) &= E(X - \mu)^2 = \sum_{\text{all } x} (x - \mu)^2 f(x) \\ &= \sum_{\text{all } x} (x^2 - 2\mu x + \mu^2) f(x) \\ &= \sum_{\text{all } x} x^2 f(x) - 2\mu \sum_{\text{all } x} x f(x) + \mu^2 \sum_{\text{all } x} f(x) \\ &= \sum_{\text{all } x} x^2 f(x) - 2\mu^2 + \mu^2 = \sum_{\text{all } x} x^2 f(x) - \mu^2 = E(X^2) - \mu^2 \end{aligned}$$

Example 3.10: Mean and variance

x	$f(x)$	$xf(x)$	$(x - \mu)^2$	$(x - \mu)^2 f(x)$
1	0.3	0.3	4	1.2
2	0.4	0.8	1	0.4
5	0.2	1.0	4	0.8
9	0.1	0.9	36	3.6
Total	1	$\mu = 3$		$\sigma^2 = 6$

Example 3.11: Alternative way to calculate the variance

x	x^2	$f(x)$	$xf(x)$	$x^2 f(x)$
1	1	0.3	0.3	0.3
2	4	0.4	0.8	1.6
5	25	0.2	1.0	5.0
9	81	0.1	0.9	8.1
Total		1	$\mu = 3$	$E(X^2) = 15$

$$\text{Var}(X) = E(X^2) - \mu^2 = 15 - 3^2 = 6$$

(4) Binomial Distribution

The binomial distribution:

n : a fixed number of independent Bernoulli trials

p : the probability of success in each trial

X : # successes in n trials

X is a binomial random variable: $X \sim \text{Bin}(n, p)$

- **Examples** in which X follows a binomial distribution:
 - Flip a coin n times. Let X = number of heads.
 - Provide a medical treatment to n subjects. Record whether each subject survives. Let X = number of survivors.
 - Ask n survey respondents: “Do you believe in capital punishment?” Let X = number who answer “Yes”.

Example 3.3 (continued)

30% of the automobiles in a certain city are foreign made.

Four cars are selected at random.

X : #cars sampled that are foreign made

F: foreign made, D: domestic

	$X = 0$	$X = 1$	$X = 2$	$X = 3$	$X = 4$
	DDDD	DDDF	DDFF	DFFF	FFFF
		DDFD	DFDF	FDFF	
		DFDD	DFFD	FFDF	
		FDDD	FDDF	FFFD	
			FDFD		
			FFDD		
Prob. of each outcome	$(1 - p)^4$	$p(1 - p)^3$	$p^2(1 - p)^2$	$p^3(1 - p)$	p^4
#outcomes	$\binom{4}{0} = 1$	$\binom{4}{1} = 4$	$\binom{4}{2} = 6$	$\binom{4}{3} = 4$	$\binom{4}{4} = 1$

- If X is binomially distributed, then

$$f(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

where $x = 0, 1, 2, \dots, n$

$$P(X = 3) = \binom{4}{3} (0.3)^3 (0.7) = 0.0756$$

$$\begin{aligned} P(X \geq 3) &= \sum_{x=3}^4 f(x) = \binom{4}{3} (0.3)^3 (0.7) + \binom{4}{4} (0.3)^4 \\ &= 0.0756 + 0.0081 = 0.0837 \end{aligned}$$

$$\begin{aligned} P(X \leq 1) &= \sum_{x=0}^1 f(x) = \binom{4}{0} (0.7)^4 + \binom{4}{1} (0.3)(0.7)^3 \\ &= 0.2401 + 0.4116 = 0.6517 \end{aligned}$$

$$P(X < 2) = P(X \leq 1) = 0.6517$$

- $\text{Bin}(n, p)$ is a probability distribution, because

$$1) f(x) \geq 0 \text{ and}$$

$$2) \sum_{x=0}^n f(x) = [p + (1 - p)]^n = 1$$

- $n = 1$: a Bernoulli trial
- Binomial tables: Table A.2

If $X \sim \text{Bin}(n, p)$, then

$$P(X \leq x) = F(x) = \sum_{k=0}^x f(k), \quad x = 0, 1, \dots, n$$

Finding binomial probabilities using R:

For $\text{Bin}(n, p)$,

- Probability distribution:
`>dbinom(x, n, p)`
- cdf:
`>pbinom(x, n, p)`

Table A.2 Binomial Distribution Table

$$P[X \leq c] = \sum_{x=0}^c \binom{n}{x} p^x (1-p)^{n-x}$$

		p										
	c	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	0.95
n = 1	0	0.950	0.900	0.800	0.700	0.600	0.500	0.400	0.300	0.200	0.100	0.050
	1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 2	0	0.903	0.810	0.640	0.490	0.360	0.250	0.160	0.090	0.040	0.010	0.003
	1	0.998	0.990	0.960	0.910	0.840	0.750	0.640	0.510	0.360	0.190	0.098
	2	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 3	0	0.857	0.729	0.512	0.343	0.216	0.125	0.064	0.027	0.008	0.001	0.000
	1	0.993	0.972	0.896	0.784	0.648	0.500	0.352	0.216	0.104	0.028	0.007
	2	1.000	0.999	0.992	0.973	0.936	0.875	0.784	0.657	0.488	0.271	0.143
	3	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 4	0	0.815	0.656	0.410	0.240	0.130	0.063	0.026	0.008	0.002	0.000	0.000
	1	0.986	0.948	0.819	0.652	0.475	0.313	0.179	0.084	0.027	0.004	0.000
	2	1.000	0.996	0.973	0.916	0.821	0.688	0.525	0.348	0.181	0.052	0.014
	3	1.000	1.000	0.998	0.992	0.974	0.938	0.870	0.760	0.590	0.344	0.185
	4	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 5	0	0.774	0.590	0.328	0.168	0.078	0.031	0.010	0.002	0.000	0.000	0.000
	1	0.977	0.919	0.737	0.528	0.337	0.188	0.087	0.031	0.007	0.000	0.000
	2	0.999	0.991	0.942	0.837	0.683	0.500	0.317	0.163	0.058	0.009	0.001
	3	1.000	1.000	0.993	0.969	0.913	0.813	0.663	0.472	0.263	0.081	0.023
	4	1.000	1.000	1.000	0.998	0.990	0.969	0.922	0.832	0.672	0.410	0.226
	5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 6	0	0.735	0.531	0.262	0.118	0.047	0.016	0.004	0.001	0.000	0.000	0.000
	1	0.967	0.886	0.655	0.420	0.233	0.109	0.041	0.011	0.002	0.000	0.000
	2	0.998	0.984	0.901	0.744	0.544	0.344	0.179	0.070	0.017	0.001	0.000
	3	1.000	0.999	0.983	0.930	0.821	0.656	0.456	0.256	0.099	0.016	0.002
	4	1.000	1.000	0.998	0.989	0.959	0.891	0.767	0.580	0.345	0.114	0.033
	5	1.000	1.000	1.000	0.999	0.996	0.984	0.953	0.882	0.738	0.469	0.265
	6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 7	0	0.698	0.478	0.210	0.082	0.028	0.008	0.002	0.000	0.000	0.000	0.000
	1	0.956	0.850	0.577	0.329	0.159	0.063	0.019	0.004	0.000	0.000	0.000
	2	0.996	0.974	0.852	0.647	0.420	0.227	0.096	0.029	0.005	0.000	0.000
	3	1.000	0.997	0.967	0.874	0.710	0.500	0.290	0.126	0.033	0.003	0.000
	4	1.000	1.000	0.995	0.971	0.904	0.773	0.580	0.353	0.148	0.026	0.004
	5	1.000	1.000	1.000	0.996	0.981	0.938	0.841	0.671	0.423	0.150	0.044
	6	1.000	1.000	1.000	1.000	0.998	0.992	0.972	0.918	0.790	0.522	0.302
	7	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

(continued)

c		p										
		0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	0.95
n = 12	0	0.540	0.282	0.069	0.014	0.002	0.000	0.000	0.000	0.000	0.000	0.000
	1	0.882	0.659	0.275	0.085	0.020	0.003	0.000	0.000	0.000	0.000	0.000
	2	0.980	0.889	0.558	0.253	0.083	0.019	0.003	0.000	0.000	0.000	0.000
	3	0.998	0.974	0.795	0.493	0.225	0.073	0.015	0.002	0.000	0.000	0.000
	4	1.000	0.996	0.927	0.724	0.438	0.194	0.057	0.009	0.001	0.000	0.000
	5	1.000	0.999	0.981	0.882	0.665	0.387	0.158	0.039	0.004	0.000	0.000
	6	1.000	1.000	0.996	0.961	0.842	0.613	0.335	0.118	0.019	0.001	0.000
	7	1.000	1.000	0.999	0.991	0.943	0.806	0.562	0.276	0.073	0.004	0.000
	8	1.000	1.000	1.000	0.998	0.985	0.927	0.775	0.507	0.205	0.026	0.002
	9	1.000	1.000	1.000	1.000	0.997	0.981	0.917	0.747	0.442	0.111	0.020
	10	1.000	1.000	1.000	1.000	1.000	0.997	0.980	0.915	0.725	0.341	0.118
	11	1.000	1.000	1.000	1.000	1.000	1.000	0.998	0.986	0.931	0.718	0.460
	12	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 13	0	0.513	0.254	0.055	0.010	0.001	0.000	0.000	0.000	0.000	0.000	0.000
	1	0.865	0.621	0.234	0.064	0.013	0.002	0.000	0.000	0.000	0.000	0.000
	2	0.975	0.866	0.502	0.202	0.058	0.011	0.001	0.000	0.000	0.000	0.000
	3	0.997	0.966	0.747	0.421	0.169	0.046	0.008	0.001	0.000	0.000	0.000
	4	1.000	0.994	0.901	0.654	0.353	0.133	0.032	0.004	0.000	0.000	0.000
	5	1.000	0.999	0.970	0.835	0.574	0.291	0.098	0.018	0.001	0.000	0.000
	6	1.000	1.000	0.993	0.938	0.771	0.500	0.229	0.062	0.007	0.000	0.000
	7	1.000	1.000	0.999	0.982	0.902	0.709	0.426	0.165	0.030	0.001	0.000
	8	1.000	1.000	1.000	0.996	0.968	0.867	0.647	0.346	0.099	0.006	0.000
	9	1.000	1.000	1.000	0.999	0.992	0.954	0.831	0.579	0.253	0.034	0.003
	10	1.000	1.000	1.000	1.000	0.999	0.989	0.942	0.798	0.498	0.134	0.025
	11	1.000	1.000	1.000	1.000	1.000	0.998	0.987	0.936	0.766	0.379	0.135
	12	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.990	0.945	0.746	0.487
	13	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 14	0	0.488	0.229	0.044	0.007	0.001	0.000	0.000	0.000	0.000	0.000	0.000
	1	0.847	0.585	0.198	0.047	0.008	0.001	0.000	0.000	0.000	0.000	0.000
	2	0.970	0.842	0.448	0.161	0.040	0.006	0.001	0.000	0.000	0.000	0.000
	3	0.996	0.956	0.698	0.355	0.124	0.029	0.004	0.000	0.000	0.000	0.000
	4	1.000	0.991	0.870	0.584	0.279	0.090	0.018	0.002	0.000	0.000	0.000
	5	1.000	0.999	0.956	0.781	0.486	0.212	0.058	0.008	0.000	0.000	0.000
	6	1.000	1.000	0.988	0.907	0.692	0.395	0.150	0.031	0.002	0.000	0.000
	7	1.000	1.000	0.998	0.969	0.850	0.605	0.308	0.093	0.012	0.000	0.000
	8	1.000	1.000	1.000	0.992	0.942	0.788	0.514	0.219	0.044	0.001	0.000
	9	1.000	1.000	1.000	0.998	0.982	0.910	0.721	0.416	0.130	0.009	0.000
	10	1.000	1.000	1.000	1.000	0.996	0.971	0.876	0.645	0.302	0.044	0.004
	11	1.000	1.000	1.000	1.000	0.999	0.994	0.960	0.839	0.552	0.158	0.030
	12	1.000	1.000	1.000	1.000	1.000	0.999	0.992	0.953	0.802	0.415	0.153
	13	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.993	0.956	0.771	0.512
	14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Contact adp@loggell.com, 863.810.671, or more information.

(continued)

		p										
	c	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	0.95
n = 15	0	0.463	0.206	0.035	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	1	0.829	0.549	0.167	0.035	0.005	0.000	0.000	0.000	0.000	0.000	0.000
	2	0.964	0.816	0.398	0.127	0.027	0.004	0.000	0.000	0.000	0.000	0.000
	3	0.995	0.944	0.648	0.297	0.091	0.018	0.002	0.000	0.000	0.000	0.000
	4	0.999	0.987	0.836	0.515	0.217	0.059	0.009	0.001	0.000	0.000	0.000
	5	1.000	0.998	0.939	0.722	0.403	0.151	0.034	0.004	0.000	0.000	0.000
	6	1.000	1.000	0.982	0.869	0.610	0.304	0.095	0.015	0.001	0.000	0.000
	7	1.000	1.000	0.996	0.950	0.787	0.500	0.213	0.050	0.004	0.000	0.000
	8	1.000	1.000	0.999	0.985	0.905	0.696	0.390	0.131	0.018	0.000	0.000
	9	1.000	1.000	1.000	0.996	0.966	0.849	0.597	0.278	0.061	0.002	0.000
	10	1.000	1.000	1.000	0.999	0.991	0.941	0.783	0.485	0.164	0.013	0.001
	11	1.000	1.000	1.000	1.000	0.998	0.982	0.909	0.703	0.352	0.056	0.005
	12	1.000	1.000	1.000	1.000	1.000	0.996	0.973	0.873	0.602	0.184	0.036
	13	1.000	1.000	1.000	1.000	1.000	1.000	0.995	0.965	0.833	0.451	0.171
	14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.995	0.965	0.794	0.537
	15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n = 16	0	0.440	0.185	0.028	0.003	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	1	0.811	0.515	0.141	0.026	0.003	0.000	0.000	0.000	0.000	0.000	0.000
	2	0.957	0.789	0.352	0.099	0.018	0.002	0.000	0.000	0.000	0.000	0.000
	3	0.993	0.932	0.598	0.246	0.065	0.011	0.001	0.000	0.000	0.000	0.000
	4	0.999	0.983	0.798	0.450	0.167	0.038	0.005	0.000	0.000	0.000	0.000
	5	1.000	0.997	0.918	0.660	0.329	0.105	0.019	0.002	0.000	0.000	0.000
	6	1.000	0.999	0.973	0.825	0.527	0.227	0.058	0.007	0.000	0.000	0.000
	7	1.000	1.000	0.993	0.926	0.716	0.402	0.142	0.026	0.001	0.000	0.000
	8	1.000	1.000	0.999	0.974	0.858	0.598	0.284	0.074	0.007	0.000	0.000
	9	1.000	1.000	1.000	0.993	0.942	0.773	0.473	0.175	0.027	0.001	0.000
	10	1.000	1.000	1.000	0.998	0.981	0.895	0.671	0.340	0.082	0.003	0.000
	11	1.000	1.000	1.000	1.000	0.995	0.962	0.833	0.550	0.202	0.017	0.001
	12	1.000	1.000	1.000	1.000	0.999	0.989	0.935	0.754	0.402	0.068	0.007
	13	1.000	1.000	1.000	1.000	1.000	0.998	0.982	0.901	0.648	0.211	0.043
	14	1.000	1.000	1.000	1.000	1.000	1.000	0.997	0.974	0.859	0.485	0.189
	15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.997	0.972	0.815	0.560
	16	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Example 3.13: $X \sim \text{Bin}(12, 0.3)$

$$(1) P(X \leq 7) = \sum_{x=0}^7 \binom{12}{x} (0.3)^x (0.7)^{12-x} = 0.9905$$

Using R, it can be obtained as

`>pbinom(7, 12, 0.3)`

$$(2) P(X = 7) = P(X \leq 7) - P(X \leq 6) = 0.9905 - 0.9614 = 0.0291$$

Using R, it can be obtained as

`>dbinom(7, 12, 0.3)`

$$(3) P(X \geq 7) = 1 - P(X < 7) = 1 - P(X \leq 6) = 1 - 0.9614 = 0.0386$$

Using R, it can be obtained as

`>1-pbinom(6, 12, 0.3)`

$$(4) P(4 \leq X \leq 7) = P(X \leq 7) - P(X \leq 3) = 0.9905 - 0.4925 = 0.4980$$

Using R, it can be obtained as

`>pbinom(7, 12, 0.3) - pbinom(3, 12, 0.3)`

$$\begin{aligned} &\text{For } X \sim \text{Bin}(n, p), \\ &\text{Expected value} = E(X) = \mu = np \\ &\text{Variance} = \text{Var}(X) = \sigma^2 = np(1 - p) \end{aligned}$$

Example 3.14: If the probability is 0.1 that a certain device fails a comprehensive safety test, what are the probabilities that among 15 of such devices,

(a) at most two will fail?

$$P(X \leq 2) = 0.8159$$

(b) at least three will fail?

$$P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.8159 = 0.1841$$

Example 3.15: Find the mean and variance of the probability distribution of the number of heads obtained in three flips of a balanced coin.

$$X \sim \text{Bin}(3, 0.5)$$

$$E(X) = \mu = np = 3(0.5) = 1.5$$

$$\text{Var}(X) = \sigma^2 = np(1 - p) = 3(0.5)(0.5) = 0.75$$

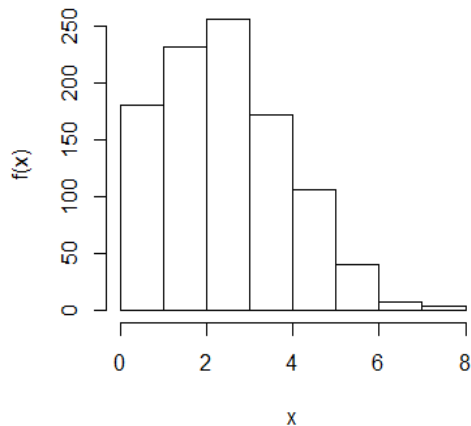
Binomial distribution

$p < 0.5$: skewed to the right

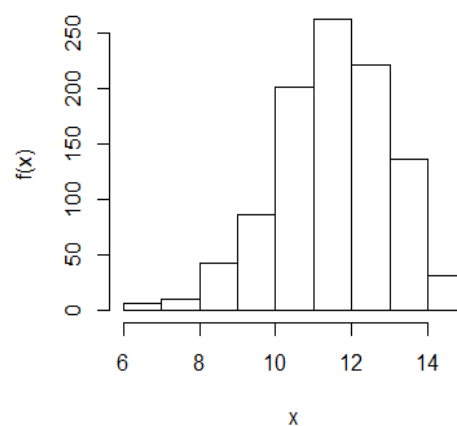
$p > 0.5$: skewed to the left

$p = 0.5$: symmetric

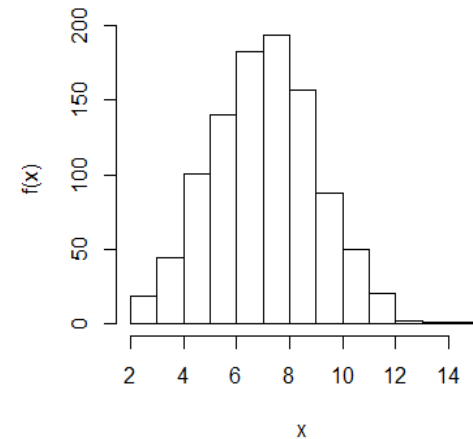
(a) Histogram of Bin(15,0.2)



(b) Histogram of Bin(15,0.8)



(c) Histogram of Bin(15,0.5)



(5) Hypergeometric Distribution

(Sampling w/o replacement)

- 1) Finite population with N individuals (binomial: infinite pop.)
- 2) S or F . There are a successes in the population.
- 3) Sample size: n . Each subset of size n is equally likely to be chosen.

$$P(X = x) = \frac{\binom{a}{x} \binom{N-a}{n-x}}{\binom{N}{n}}$$

$$\max\{0, n - N + a\} \leq x \leq \min\{n, a\} \quad (\because x \leq a \text{ and } n - x \leq N - a)$$

Using R,

- Probability distribution:
`>dhyper(x, a, N-a, n)`
- cdf:
`>phyper(x, a, N-a, n)`

Example 3.17: A shipment of 25 CDs contains 5 that are defective. If 10 of them are randomly chosen without replacement, what is the probability that 2 of the 10 will be defective?

$$x = 2, n = 10, a = 5, N = 25$$

$$f(2) = P(X = 2) = \frac{\binom{5}{2}\binom{20}{8}}{\binom{25}{10}} = \frac{10 \cdot 1,511,640}{49,031,400} = 0.385$$

Using R,

```
>dhyper(2, 5, 20, 10)
```

Hypergeometric Distribution

$$E(X) = \frac{na}{N}$$
$$Var(X) = \frac{na(N-a)(N-n)}{N^2(N-1)}$$

Let $\frac{a}{N} = p$. Then $E(X) = np$

$$Var(X) = np(1-p) \left(\frac{N-n}{N-1} \right)$$

If $N \gg n$, then $Var(X) \approx np(1-p)$

$\frac{N-n}{N-1}$: finite population correction factor

- Binomial approximation to the hypergeometric, when $\frac{n}{N} \leq 0.1$, and $p = \frac{a}{N}$ is not too close to either 0 or 1.

Example 3.18: From Example 3.17, for a lot of 100 CDs, 20 are defective . Find the probability that among a randomly selected sample of 10 CDs, 2 are defective, by using
(a) The formula for the hypergeometric distribution:

$$x = 2, n = 10, a = 20, N = 100$$
$$f(2) = \frac{\binom{20}{2} \binom{80}{8}}{\binom{100}{10}} = 0.318$$

>dhyper(2, 20, 80, 10)

(b) The binomial approximation to the hypergeometric:

$$x = 2, n = 10, p = \frac{a}{N} = \frac{20}{100} = 0.2, \frac{n}{N} = 0.1$$

$$f(x) = \binom{10}{2} (0.2)^2 (0.8)^8 = 0.302 \text{ (close)}$$

>dbinom(2, 10, 0.2)

(6) Poisson Distribution

- Properties:
 1. The probability that an event occurs in the interval is proportional to the length of the interval.
 2. Two events cannot occur at exactly the same instant.
 3. Events occur independently.
- The probability of x events occurring in a time period t for a Poisson random variable with parameter λ is

$$P(X = x) = f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, x = 0, 1, 2, \dots \quad (e=2.71828)$$

$$E(X) = Var(X) = \lambda$$

Poisson Distribution

Maclaurin Series:

$$e^{\lambda} = 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \cdots = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$$

$$\sum f(x) = \sum \frac{\lambda^x e^{-\lambda}}{x!} = e^{-\lambda} e^{\lambda} = 1$$

Therefore, $f(x)$ is a probability distribution.

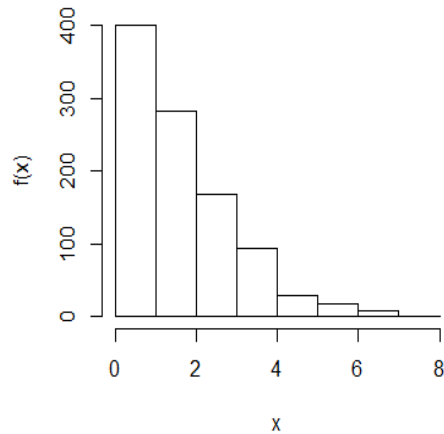
- Use of Poisson distributions, examples:
 - number of occurrences in a given time
 - distribution of bomb hits in an area
 - distribution of fish in a lake
- Poisson tables: Table A.3

$$P(X \leq x) = F(x) = \sum_{k=0}^x f(k)$$

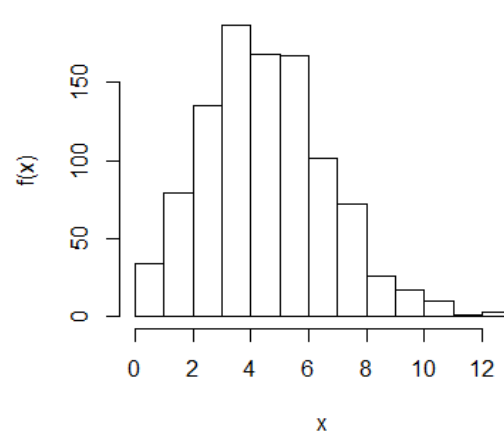
Poisson Distribution

- The Poisson distribution is highly skewed for small values of λ .
- As λ increases, the distribution becomes more symmetric.

(a) Histogram of Poisson(2)



(b) Histogram of Poisson(5)



(c) Histogram of Poisson(10)

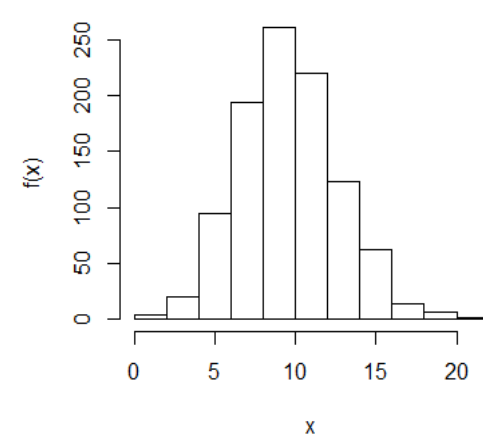


Table A.3 Poisson Distribution Table $P(X \leq x)$

$\lambda =$	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.2	1.4	1.6	1.8
$x = 0$	0.9048	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966	0.4493	0.4066	0.3679	0.3012	0.2466	0.2019	0.1653
1	0.9953	0.9825	0.9631	0.9384	0.9098	0.8781	0.8442	0.8088	0.7725	0.7358	0.6626	0.5918	0.5249	0.4628
2	0.9998	0.9989	0.9964	0.9921	0.9856	0.9769	0.9659	0.9526	0.9371	0.9197	0.8795	0.8335	0.7834	0.7306
3	1.0000	0.9999	0.9997	0.9992	0.9982	0.9966	0.9942	0.9909	0.9865	0.9810	0.9662	0.9463	0.9212	0.8913
4	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9992	0.9986	0.9977	0.9963	0.9923	0.9857	0.9763	0.9636
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9994	0.9985	0.9968	0.9940	0.9896
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9994	0.9987	0.9974
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9994
8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$\lambda =$	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8	4.0	4.5	5.0	5.5
$x = 0$	0.1353	0.1108	0.0907	0.0743	0.0608	0.0498	0.0408	0.0334	0.0273	0.0224	0.0183	0.0111	0.0067	0.0041
1	0.4060	0.3546	0.3084	0.2674	0.2311	0.1991	0.1712	0.1468	0.1257	0.1074	0.0916	0.0611	0.0404	0.0266
2	0.6767	0.6227	0.5697	0.5184	0.4695	0.4232	0.3799	0.3397	0.3027	0.2689	0.2381	0.1736	0.1247	0.0884
3	0.8571	0.8194	0.7787	0.7360	0.6919	0.6472	0.6025	0.5584	0.5152	0.4735	0.4335	0.3423	0.2650	0.2017
4	0.9473	0.9275	0.9041	0.8774	0.8477	0.8153	0.7806	0.7442	0.7064	0.6678	0.6288	0.5321	0.4405	0.3575
5	0.9834	0.9751	0.9643	0.9510	0.9349	0.9161	0.8946	0.8705	0.8441	0.8156	0.7851	0.7029	0.6160	0.5289
6	0.9955	0.9925	0.9884	0.9828	0.9756	0.9665	0.9554	0.9421	0.9267	0.9091	0.8893	0.8311	0.7622	0.6860
7	0.9989	0.9980	0.9967	0.9947	0.9919	0.9881	0.9832	0.9769	0.9692	0.9599	0.9489	0.9134	0.8666	0.8095
8	0.9998	0.9995	0.9991	0.9985	0.9976	0.9962	0.9943	0.9917	0.9883	0.9840	0.9786	0.9597	0.9319	0.8944
9	1.0000	0.9999	0.9998	0.9996	0.9993	0.9989	0.9982	0.9973	0.9960	0.9942	0.9919	0.9829	0.9682	0.9462
10	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9995	0.9992	0.9987	0.9981	0.9972	0.9933	0.9863	0.9747
11	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9996	0.9994	0.9991	0.9976	0.9945	0.9890
12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997	0.9992	0.9980	0.9955

Finding Poisson Probabilities Using R

For $\text{Poisson}(\lambda)$,

- Probability distribution:
`>dpois(x,  $\lambda$ )`
- cdf:
`>ppois(x,  $\lambda$ )`

Example 3.19: A 911 operator handles 4 calls every 3 hours on average.

(a) What is the probability of no calls in the next hour?

$$E(X) = \lambda = 4/3$$

Let X be the number of calls in an hour. Then $X \sim \text{Poisson}(4/3)$.

$$f(0) = \frac{(4/3)^0 e^{-4/3}}{0!} = e^{-4/3} = 0.264$$

>dpois(0,4/3)

(b) Find the probability of at most two calls in the next hour.

$$\begin{aligned} P(X \leq 2) &= \sum_{x=0}^2 f(x) = \frac{(4/3)^0 e^{-4/3}}{0!} + \frac{(4/3)^1 e^{-4/3}}{1!} + \frac{(4/3)^2 e^{-4/3}}{2!} \\ &= e^{-4/3} \left(1 + \frac{4}{3} + \frac{16}{18} \right) = \frac{29}{9} e^{-4/3} = 0.849 \end{aligned}$$

>ppois(2, 4/3)

Example 3.20:

On average, 12 cars pass a toll gate booth in a minute during rush hours.

(a) Probability that one car passes the booth in 3 seconds:

$$\lambda = (0.2)3 = 0.6$$

$$f(1) = F(1) - F(0) = 0.878 - 0.549 = 0.329$$

>dpois(1, 0.6)

(b) Probability that at least two cars pass in 5 seconds:

$$\lambda = (0.2)5 = 1$$

$$P(X \geq 2) = 1 - P(X \leq 1) = 1 - F(1) = 1 - 0.736 = 0.264$$

>1-ppois(1, 1)

(c) Probability that at most one car passes in 10 seconds:

$$\lambda = (0.2)10 = 2$$

$$P(X \leq 1) = 0.406$$

>ppois(1, 2)

Poisson approximation to the binomial

Let $X \sim \text{Bin}(n, p)$.

Let $n \rightarrow \infty$ and $p \rightarrow 0$ such that np remains constant (np is a moderate number). Then

X is approximately $\text{Poisson}(\lambda)$ such that $\lambda = np$.

$n \geq 20, p \leq 0.05$: acceptable

$n \geq 100, p \leq 0.01$: excellent approximation

$$X \sim \text{Bin}(n, p) \implies E(X) = np = \lambda$$

$$\text{Var}(X) = np(1 - p) = \lambda(1 - p) \rightarrow \lambda$$

Example 3.21: A publisher of mystery novels tries to keep its books free of typos. The probability of any given page containing at least one typo is 0.003 and errors are independent from page to page. What is the approximate probability that a 500 page book has

(a) exactly 1 page with typos?

If X is the number of pages with typos, then $X \sim \text{Bin}(n, p)$.

$$n = 500, p = 0.003 \Rightarrow \lambda = np = (500)(0.003) = 3/2$$

$$P(X = 1) \approx \frac{(3/2)e^{-3/2}}{1!} = 0.3347$$

>dpois(1, 1.5)

(b) at most 2 pages with typos?

$$P(X \leq 2) \approx \sum_{x=0}^2 \frac{1.5^x e^{-1.5}}{x!} = \left(1 + \frac{3}{2} + \frac{(3/2)^2}{2}\right) e^{-1.5} = \frac{29}{8} e^{-1.5} = 0.809$$

>ppois(2, 1.5)

(7) Geometric Distribution

1. Each trial: Success (S) or Failure (F)
2. Independent trials
3. $P(S) = p$
4. Continues until the first success is observed

- Probability of getting the first success on the x -th trial:

$$f(x) = p(1 - p)^{x-1}, x = 1, 2, \dots$$

$$E(X) = \frac{1}{p}$$

- Using R,
probability distribution
`>dgeom(x-1, p)`
cdf
`>pgeom(x-1, p)`

Example 3.23: A fair die is tossed until a certain number appears. What is the probability that the first six appears at the fifth toss?

$$x = 5, p = 1/6$$

$$P(X = 5) = (1/6)(5/6)^{5-1} = 0.080$$

Using R,
>dgeom(4, 1/6)

(8) Chebyshev's Inequality

If a probability distribution has mean μ and sd σ , the probability of deviation from μ by at least $k\sigma$ is at most $1/k^2$.

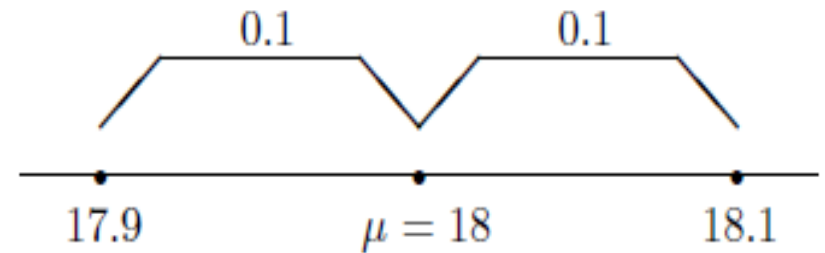
$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Equivalently,

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

Example 3.24: The bearings made by a certain process have a mean radius of 18 mm with a standard deviation of 0.025 mm. With what minimum probability can we assert that the radius of a bearing is between 17.9 mm and 18.1 mm?

$$17.9 = \mu - 0.1, 18.1 = \mu + 0.1 \Rightarrow |X - \mu| \leq 0.1$$



$$0.1 = k\sigma = 0.025k \Rightarrow k = \frac{0.1}{0.025} = 4$$

$$P(|X - \mu| \geq 0.1) = P(|X - \mu| \geq 4\sigma) \leq \frac{1}{4^2} = \frac{1}{16}$$

$$P(|X - \mu| < 0.1) = 1 - P(|X - \mu| \geq 0.1) \geq 1 - \frac{1}{16} = \frac{15}{16}$$

At least 15/16

Example 3.25: What can you say about the probability that a random variable falls within two standard deviations of its mean?

$$P(-2\sigma < X - \mu < 2\sigma) = P(|X - \mu| < 2\sigma) \geq 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4}$$

(9) Multinomial Distribution

Each trial results in any one of k possible outcomes.

$$p_i = P(\text{outcome } i \text{ on any trial})$$

X_i : #trials resulting in outcome i

$$f(x_1, \dots, x_k) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k},$$

$$x_i = 0, 1, \dots, n, x_1 + \cdots + x_k = n$$

Example 3.26: Toss a drum 10 times.

Let X_1 : #heads, X_2 : #sides, X_3 : #tails.

Let $n = 10$, $p_1 = \frac{1}{4}$, $p_2 = \frac{1}{2}$, $p_3 = \frac{1}{4}$. Then

$$P(X_1 = 2, X_2 = 5, X_3 = 3) = \frac{10!}{2!5!3!} (1/4)^2 (1/2)^5 (1/4)^3 = 0.0769$$